Audit of the SV efficiency proof, round 8: the merged article

Source audited: the merged article at commit 58764bb. That is the author's revised `sv_efficiency_proof.tex` (revisions marked `\rev`, blue) with the uniform-versus-adaptive additions in `numeric.tex` (marked `\adapt`, magenta). Earlier rounds: 1–7. Not refereed.

Part A. Proof-check

A1. Scope

This round checks:

- every passage marked `\rev`: (H1), (H3), (H6), Remark 2.1, Remark 2.3, the σ = 0 convention of §3, Theorem 4.2 with $C_{\rm loc}/C_{\rm glob}$, Assumption 5.5, Lemma 5.11(i), Remark 6.1, the contrast remark of §9, and the bibliography;
- every passage marked `\adapt` (§10.9 and four sentences), against `output/aconv.json` and `output/asweep.json`;
- the consistency of the revisions with the rest of the article and with the code.

A2. Step table

	check	status		
	used by Friedrichs/Poincaré and by the Neumann face data	OK; makes explicit what was used tacitly		
	κ is monotone in c_b, so any upper bound of $\|c-\frac12\mathrm{div}\,b\|_\infty$ gives a valid, larger κ	OK. The code takes no such input (**W1**)		
	§8 uses the A of the iterate	OK		
	$(A\nabla u,\nabla v)=\sum_K\sum_{E\subset K}\frac{\beta_E}{n_E}(\nabla u\cdot t_E)(\nabla v\cdot t_E)=\sum_E\beta_E(u_j-u_i)(v_j-v_i)$, since $\sum_{K\supset E}1/n_E=1$	OK (re-derived); the scope sentence is correct		
ostic"	finite sampling does not certify a supremum	OK. §10.4 and the caption of Table 3 still call the computed c_b "the upper bound" (**W2**)		
	matches the estimator and Theorem 4.1	OK		
end on	Lemma 5.11(ii) is the only place C_Π enters	OK. The abstract and contribution 2 of §1 still say "the constants depend only on d and the shape regularity" (**W3**)		
d	the CW08/Licht constants depend on the domain	OK		
	c_a	$	sign of $\ell(\psi_a)$	OK (a genu
	split local/global constants	OK		
	see A4	**W4**		
	arXiv metadata of the revised entries	OK, but four entries are no longer cited: BV00, Pe02, He25, He26 (**W5**)		
	every number against the JSON data; no run has I, $I_\oplus$, I_S, $I_E<1$	OK		

A3. Verdict

- **The mathematics is correct**, given A2 (Neumann version, [EV15]) and Assumption 5.5. The revisions are improvements; Remark 2.1 and Lemma 5.11(i) close two small gaps.
- **W1** is a code gap: the implementation cannot take the certified bounds that the revised (H3) requires.
- **W2–W5** are text consistency.

A4. Findings

W1 (code). The revised (H3) asks for certified input bounds of σ and of $|c-\frac12\mathrm{div}\,b|_\infty$. `abr17_estimator` / `sv_estimator` take σ, but no upper bound of the supremum; they always use the sampled enclosure of `reaction_cell_bounds`.

- **Fix:** an optional argument `cb_sup` (a certified upper bound of $|c-\frac12\mathrm{div}\,b|_\infty$). When given, it is used in place of the sampled enclosure and the "not strictly guaranteed" note is dropped. When it is absent, the current diagnostic behaviour stays.

W2 (text, magenta and Table 3). §10.4 says "the computed c_b is the upper bound of Remark 2.3", and the caption of Table 3 says "computed upper bounds". After the revision of Remark 2.3 these are sampled enclosures, diagnostic only.

- **Fix:** "the computed c_b is the sampled enclosure of Remark 2.3 (diagnostic); here it exceeds the true supremum $c_{\max}$ by 3–12 %".

W3 (text). The abstract says "The constants depend only on the space dimension and the shape regularity of the mesh", and contribution 2 of §1 says the same. Theorem 4.2 now allows $C_{\rm glob}$ to depend on C_Π.

- **Fix:** "The local constants depend only on …; the global constant may also depend on the constant of the commuting projection (Assumption 5.5), but on none of the problem parameters."

W4 (the contrast remark). The revised sentence reads: "a contrast-independent local efficiency constant here would require a change in the scope or structure of the estimator, not merely an appeal to quasi-monotonicity". This is not right as stated, and a sharper statement is available.

- **What is known.** For conforming elements, robust efficiency in the jump holds for the weighted residual estimator of Bernardi–Verfürth; uniform reliability needs quasi-monotonicity ([BV00], [Pe02]; see the summary in Cai–He–Zhang, arXiv 1510.06481). Li's lower bound [Li26] is built on a checkerboard cross-point, which is **not** quasi-monotone. Restricting to quasi-monotone ε is therefore exactly the classical remedy. It is a change of scope, but "not merely an appeal to quasi-monotonicity" suggests the opposite.
- **The sharper statement.** Our bound is guaranteed with reliability constant 1 for every contrast (Theorem 9.2, part 1). Li's theorem bounds the product $C_{\rm rel}C_{\rm eff}\gtrsim M^{1/4}$ for every finite-range local estimator. It therefore implies directly that **the efficiency constant of this estimator must grow at least like $M^{1/4}$ for checkerboard coefficients**: the contrast dependence in Theorem 9.2, part 2, cannot be removed in general. Under quasi-monotonicity a contrast-independent efficiency constant is not excluded, but it is not proved here.
- **Fix** (replacing the sentence):

> Since the bound of Theorem 9.2 is guaranteed, with reliability constant 1 for every contrast, the result of \cite{Li26} implies that its efficiency constant must grow at least like $M^{1/4}$ for checkerboard coefficients; the contrast dependence in part 2 cannot be removed in general. For quasi-monotone coefficients, robust bounds are known for residual estimators \cite{BV00,Pe02}; whether the present estimator is robustly efficient in that class is open.

This also restores the citations of BV00 and Pe02 (W5).

W5 (bibliography). BV00, Pe02, He25 and He26 are listed but not cited. W4's fix cites BV00 and Pe02.

- **For He25/He26**, either remove them or cite them in the contrast remark: He26 (arXiv 2601.01585) claims a robust framework "for both DG and conforming discretizations", which must be reconciled with [Li26]. The likely reconciliation is robustness of the recovered-flux error, or a non-finite-range step. This is the open item from round 4, still unread in full.

A5. Optimality

- **The revisions remove tacit hypotheses** (connectedness, mesh-aligned boundary partition, the dependence of C_Π on the domain). The hypothesis lists are now complete and still minimal, except for M1 of round 7: Theorem 4.2 needs neither the inflow part of (H4) nor $b\in W^{1,\infty}$.
- **New sharpness statement (from W4):** for variable diffusion, the contrast dependence of the efficiency constant is necessary in general, with the rate $M^{1/4}$ as a lower bound. This turns an open limitation into a proved one and is worth stating.

- **The cellwise-σ track** remains the most useful extension. It would remove c_b, the Theorem 7.2 restriction, μ, and with them the need for certified bounds of $|c-\frac12\mathrm{div}\,b|_\infty$ (W1).

Part B. Literature scan (round 8)

B1. Answer

No change in novelty or track. The 2026 searches turned up equilibrated-flux work for interface problems, both CutFEM and conforming/DG. Nothing combines convection, stabilisation and an augmented norm. The new items bear only on the variable-diffusion remark (W4).

B2. Frontier additions

result	scheme	relevance
Cai, He, Zhang, arXiv 1510.06481	conforming, nonconforming and DG; interface problems	conforming residual estimator: robust efficiency, reliability robust only under quasi-monotonicity; nonconforming/DG: robust without it. Supports W4
Capatina, Gouasmi, arXiv 2604.02137 (2026) and arXiv 2507.06740	CutFEM, elliptic interface	equilibrated fluxes with discontinuous coefficients; no convection [abstract]
He, arXiv 2601.01585 (2026)	DG and conforming, interface	claims robustness for conforming as well; reconcile with [Li26] (W5) [abstract]

B3. Open items, ranked by novelty-to-effort

1. **W2–W5:** text only.
2. **W1:** the `cb_sup` input. Low effort.
3. **Read He26 against Li26.** Low effort; settles W5 and the contrast remark.
4. **The cellwise-σ track.** Medium effort.
5. **Robust efficiency under quasi-monotonicity for the present estimator.** Open, medium–high; the natural follow-up to W4.
6. **A5 (Licht 2019, Theorem 1); the stopped iterate against the discrete solution.** As before.

B4. References (new in round 8)

- Z. Cai, C. He, S. Zhang, Finite element methods for interface problems: robust residual-based a posteriori error estimates, arXiv 1510.06481. [abstract and summary]
- D. Capatina, A. Gouasmi, arXiv 2604.02137 and arXiv 2507.06740. [abstract]
- Purdue dissertation of C. He, record. [abstract]
- Berrone, M2AN 40 (2006), numdam (uses quasi-monotonicity for jump-robust bounds). [abstract]